Individual presentation recording

Start Assignment

- Due 17 May by 23:59
- Points 10
- Submitting a file upload
- File types mp4

Invidividual presentation recording instructions

Reading research papers and presenting research work is an essential skill to develop for both AI research and industry.

For this assignment:

- You need to select a research paper among a provided list (see below). Register your choice before **Sunday 29th of March**. After this date, you will be assigned randomly to an available paper.
- You **must present the paper you originally chose or were assigned to**. *Any failure to follow this rule will lead to your presentation not being graded*.
- Youy will record a video in which you present the selected research paper. The video should be **6 min long +- 30 sec**.
- Your presentation will be marked based on the submitted video.
- The top 6 or 7 videos will be selected and their authors will be invited to present live in the last session of the semester (Thursday of week 12). This live presentations will not influence the grades, but will be used as a showcase of the best presentations.
- The papers presented in the live presentations will be examinable in the Final Exam.

The following table contains links to the papers. You need to choose one paper (**max. 8 students per paper**) and register your choice by **signing-up to the corresponding group (https://canvas.auckland.ac.nz/courses/141895/groups)** (first come, first served).

List of papers for video presentations

Link to paper
Logic Theorist (https://canvas.auckland.ac.nz/courses/141895/files/folder/Papers?preview=16939199)
Alpha Fold (https://www.nature.com/articles/s41586-021-03819-2.pdf)
Inverse reinforcement learning (http://proceedings.mlr.press/v48/finn16.pdf)
FunSearch (https://www.nature.com/articles/s41586-023-06924-6.pdf)
Large language model-powered robots (https://openreview.net/pdf?id=wMpOMO0Ss7a)
Probabilistic logical neural networks (https://proceedings.neurips.cc/paper_files/paper/2019/file/13e5ebb0fa112fe1b31a10
Fuzzy Logic flight control (https://canvas.auckland.ac.nz/courses/141895/files/16939200?wrap=1) (https://canvas.auckland.ac.nz/courses/141895/files/16939200/download?download_frd=1)
Bayesian malware detection (https://canvas.auckland.ac.nz/courses/141895/files/16939202?wrap=1) (https://canvas.auckland.ac.nz/courses/141895/files/16939202/download?download_frd=1)
NEAT Pacman (https://canvas.auckland.ac.nz/courses/141895/files/16939201?wrap=1) (https://canvas.auckland.ac.nz/cours download_frd=1)
Layered control for a robot (https://canvas.auckland.ac.nz/courses/141895/files/16939178?wrap=1) (https://canvas.auckland.ac.nz/courses/141895/files/16939178/download?download_frd=1)
Complex flocking dynamics (https://canvas.auckland.ac.nz/courses/141895/files/16939203?wrap=1) (https://canvas.auckland.ac.nz/courses/141895/files/16939203/download?download_frd=1)

Robot soccer (https://canvas.auckland.ac.nz/courses/141895/files/16939176?wrap=1) (https://canvas.auckland.ac.nz/course download_frd=1)
Cognitively Inspired Learning of Incremental Drifting Concepts (https://arxiv.org/pdf/2110.04662)
InstructGPT (https://arxiv.org/abs/2203.02155)
Automatic prompt optimization with GD and beam search (https://arxiv.org/abs/2305.03495)
Deep Generative Model for Nowcasting (https://www.nature.com/articles/s41586-021-03854-z)
Sanity Checks for Saliency Maps (https://arxiv.org/pdf/1810.03292)
Continual Learning: Applications and the Road Forward (https://arxiv.org/pdf/2311.11908.pdf)
Visual Programming: Compositional visual reasoning (https://openaccess.thecvf.com/content/CVPR2023/papers/Gupta_Visual_Programming_Compositional_Visual_Reasoning_With
Chain-of-Thought Reasoning (https://openreview.net/pdf?id=_VjQlMeSB_J)

The papers are releated to the topics covered in the course. However, you are highly likely to encounter concepts that are not covered in the lectures. Indeed, it is common to encounter new concepts when reading research papers, even in a familiar field of study. You will need to do some personal research to understand them. Do not hesitate to read related papers, blog articles, or even to use GenAI tools to explore and better understand the context in which the paper you choose sits in.

Knowing this, we highly rercommend you do not wait last minute to read the paper and prepare your presentation. A good presentation takes some time to prepare!

Deliverable and timing

Deliverable: a 6-minute (+- 30sec) **video file in format .mp4** (make sure the video plays and is not corrupted if you convert it from another format).

Deadline: you must submit your video by **Sunday 17th of May at 23:59pm** on this Canvas page.

- **Late submission penalties:**
  - Up to 24h late: −10%
  - 24–48h late: −20%
  - More than 48h late: −100%

The presentation
6 minutes is not a lot of time – focus on a few succinct messages you deliver clearly!

Here is the structure we are expecting for the presentations, with some general ideas of what you might want to discuss (not all the points might be relevant to your paper) :

1. Start by presenting the context and motivation of the paper. You will likely want to introduce some baseline concepts coming from your personal research on the topic, as not everyone will be familiar with the theme of the paper.
2. Explain clearly what research gap was identified by the authors and what research question(s) they are trying to answer, or what problem they are trying to solve, i.e., why is this an interesting and important piece of work? This part should include some references to related work.
3. Explain what was done by the authors to answer the identified research gap, including what methodology was developed/used, and how it was evaluated. You might want to explain how it extended related work where relevant, e.g., how it helps solve identified limitations.
4. Discuss the main results and how they support the contributions made by the authors (e.g., showing a gain of performance, or also illustrating limitations).

5. Present the conclusions and future work directions arising from this research. This can come directly from the authors, but you can also add your own thoughts!

Feel free to use figures from the paper if helpful (without forgetting to reference them!), or to create new visuals to illustrate key concepts or features (doing the latter may lead to a better looking visual and one that is easier to work within the available time).

If you use figures from the paper or another paper without referencing them, a **0.5 mark penalty** will be applied.

Remember that this is a scientific presentation. The goal is to explain what work was done and how it advances knsoweledge in the field, and not sell a product.

Use of GenAI

Feel free to get help from GenAI to get a better understanding of the paper, e.g., by asking specific questions about some content you struggle to understand, or releted references to check out. You can also generate ideas to help you structure your presentation and create visuals. We also recommend you to create your slides yourself, even if you can use GenAI to help a little.

However, **you are not allowed to directly generate the video via an AI tool, or use an AI generated voice**. You must be the presenter! The goal of the exercise is for you to develop skills at presenting a scientific paper in English, so there is no point in generating the full thing with AI.

Papers allocation (TBA)

Papers allocatio

Paper
Logic Theorist (https://canvas.auckland.ac.nz/courses/141895/files/16939199?wrap=1) (https://canvas.auckland.ac.nz/courses/141895/files/16939199/download?download_frd=1)
Alpha Fold (https://www.nature.com/articles/s41586-021-03819-2.pdf)
Inverse reinforcement learning (http://proceedings.mlr.press/v48/finn16.pdf)
FunSearch (https://www.nature.com/articles/s41586-023-06924-6.pdf)
Large language model-powered robots (https://openreview.net/pdf?id=wMpOMO0Ss7
Probabilistic logical neural networks (https://proceedings.neurips.cc/paper_files/paper/2019/file/13e5ebb0fa112fe1
Fuzzy Logic flight control (https://proceedings.neurips.cc/paper_files/paper/2019/file/13e5ebb0fa112fe1b31a1
Bayesian malware detection (https://canvas.auckland.ac.nz/courses/141895/files/16939202?wr
NEAT Pacman (https://canvas.auckland.ac.nz/courses/141895/files/16939201?wrap=1)
Layered control for a robot (https://canvas.auckland.ac.nz/courses/141895/files/16939178?wra
Complex flocking dynamics (https://canvas.auckland.ac.nz/courses/141895/files/16939203?wr
Robot soccer (https://canvas.auckland.ac.nz/courses/141895/files/16939176?wrap=1)
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Deep Generative Model for Nowcasting (https://www.nature.com/articles/s41586-021-038
Sanity Checks for Saliency Maps (https://papers.nips.cc/paper_files/paper/2018/file/294a8ed24b1ad22ec2e7e
Continual Learning: Applications and the Road Forward (https://arxiv.org/pdf/2311.1190

Visual Programming: Compositional visual reasoning (https://openaccess.thecvf.com/content/CVPR2023/papers/Gupta_Visual_Programming_Compositional_Visual_Reasoning_With

Chain-of-Thought Reasoning (https://openreview.net/pdf?id=_VjQlMeSB_J)

Presentation rubric

Criteria	Ratings					Pts
Criterion 1: Content Quality and Understanding	**5 Pts** **Excellent** Clear, accurate, and concise coverage of the paper. Effectively explains context, motivation, research gap, methodology, results, conclusions and future work. Demonstrates strong understanding and good prioritisation within the 6-minute limit.	**3.75 Pts** **Good** Covers most key aspects with good understanding. Minor omissions or uneven depth, but overall coherent and relevant.	**2.5 Pts** **Satisfactory** Main ideas are presented, but explanation lacks depth, clarity, or balance. Some important aspects are missing or underdeveloped.	**1.25 Pts** **Limited** Incomplete or unclear explanation. Significant gaps in understanding or missing key components.	**0 Pts** **Poor** Little or no understanding of the paper. Content is incorrect, incoherent, or off topic.	5 pts
Criterion 2: Presentation Style and Delivery	**3 Pts** **Excellent** Clear, confident, and engaging delivery. Logical structure, good pacing, and effective communication.	**2.25 Pts** **Good** Generally clear and well-structured. Minor issues with pacing, tone, or engagement.	**1.5 Pts** **Satisfactory** Main ideas are presented, but explanation lacks depth, clarity, or balance. Some important aspects are missing or underdeveloped.	**0.75 Pts** **Limited** Incomplete or unclear explanation. Significant gaps in understanding or missing key components.	**0 Pts** **Poor** Little or no understanding of the paper. Content is incorrect, incoherent, or off topic.	3 pts
Criterion 3: Visual / Presentation Support	**2 Pts** **Excellent** Excellent High-quality visuals that are clear, readable, well-designed, and well-aligned with the spoken content.	**1.5 Pts** **Good** Visuals are clear and helpful, with minor readability or design issues.	**1 Pts** **Satisfactory** Visuals are present but cluttered, hard to read, or only loosely connected to the talk.	**0.5 Pts** **Limited** Visual support is minimal, poorly designed, difficult to read, or inconsistently used, limiting its usefulness.	**0 Pts** **Poor** Visual support is missing or seriously detracts from the presentation.	2 pts
						Total points: 10